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The Clouclip, a wearable device for measuring near-work and outdoor time: validation and comparison of objective measures with questionnaire estimates

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ABSTRACT.

Purpose: **To validate a novel wearable device that can measure both viewing distance and light exposure, Clouclip, and compare questionnaire estimates regarding near-work and outdoor time with the objective measures obtained using Clouclip.**

Methods: **Fifteen Clouclips were selected to measure different distances and levels of illuminance. With each Clouclip, five measurements at different distances and light intensities were measured and recorded. Eighty participants wore Clouclips for a week and completed an activity questionnaire afterwards.**

Results: **The intra- and inter-Clouclip coefficients were 1.00 and 0.99 for measuring distance and 1.00 and 1.00 for illuminance, respectively. Within the measurement limit, the maximum relative error was 2.07% for distance and 2.23% for illuminance. Assuming that <30 cm was the typical distance for near-work activities and >1000 Lux was the typical cut-off for outdoor environments, the questionnaire showed a trend of overestimation for both. The greatest overestimation of near-work occurred during the school period [Questionnaire: 4.73 hr (4.73, 5.07) versus Clouclip: 2.16 hr (1.74, 2.78); $p < 0.01$]. The greatest overestimation of outdoor activity also occurred during the school period [Questionnaire: 1.60 hr (1.33, 1.85) versus Clouclip: 1.21 hr (0.96, 1.50); $p < 0.01$]. Based on Clouclip, the total time spent outdoors was estimated to be 1.55 hr on school days, of which 0.34 hr occurred after school. For weekend days, however, the duration was only 0.17 hr.**

Conclusions: **Clouclip had excellent precision and accuracy. Although the agreement between the questionnaire and Clouclip was relatively poor, they were able to complement each other to provide a more logical and feasible assessment of exposure to near-work and outdoor activity. Indoor-oriented lifestyles were found to predominate in Chinese children.**

Key words: myopia – near work – outdoor exposure – questionnaire – wearable device

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Introduction

The key to preventing myopia is the identification of modifiable risk factors for myopia. One major risk factor that has been generally accepted is lifestyle changes (Morgan et al. 2012). Of the many features relevant to lifestyle, two major features that have been widely investigated are the time spent in near-work and time spent outdoors (Morgan et al. 2012). A meta-analysis by Huang et al. (2015) summarized that, regardless of some inconsistencies, the evidence tended to suggest that more time spent on near-work activities was associated with increased chances of developing myopia. In contrast, the association between time outdoors and myopia has been a relatively consistent finding in the literature (Xiong et al. 2017). Nonetheless, the underlying mechanism is still not completely clear (Torii et al. 2017).

A major challenge has been the development of methods to objectively measure near-work and time outdoors. Most studies have been carried out using questionnaires, which are associated with recall bias and inaccuracy (Jones-Jordan et al. 2012). Many questionnaires have also asked for estimates outside of school hours, assuming that experiences would be quite standard during school hours. Given the lack of

a well-accepted definition of near work, different types of activities have been defined as near-work. For instance, some have only included reading (Saw et al. 2002), while others have included playing with pets, playing chess and cards, and even cooking (Li et al. 2015). The parameters to quantify the workload of near-work also differs remarkably between studies. Some only utilize the total duration of the activities (Loman et al. 2002), while others employed a 'dioptre-hour' to reflect the intensity of near-work (Mutti et al. 2002). Recently, researchers have started to consider the impact of the temporal pattern of an activity on the intensity (i.e. continuous versus intermittent near work in cases where the total duration is identical) (Wen et al. 2020). In terms of outdoor activities, earlier studies have identified time spent outdoors, which was mixed with sports, as a protective factor inhibiting the development of myopia (Parssinen & Lyyra 1993; Mutti et al. 2002; Jones et al. 2007). However, a more recent study, which employed a more detailed questionnaire in which physical activities were separated from time outdoors, showed that it was the time outdoors and not the sport, per se, that manifested in a protective effect (Rose et al. 2008a).

Owing to the development of portable and wearable devices, objective measurements of time outdoors and light exposure have been obtained using devices embedded with a light sensor, such as HOBO (Onset Computer Corporation, Buzzards Bay, MA, USA) (Dharani et al. 2012; Schmid et al. 2013), Actiwatch (Philips Respironics, Inc., Pittsburgh, PA, USA) (Read et al. 2015; Ostrin, 2017), and Fitsight (Coinventors: SSM, SHP, ZXQ, Singapore) (Verkicharla et al. 2017; Read et al. 2018). Using a distance sensor, Leung et al. (2011) developed an instrument that could measure viewing distance. Ostrin et al. also developed a similar device that had reliable precision and accuracy (Williams et al. 2019). However, Clouclip has both a light and a distance sensor, making it the first that can measure both light intensity and distance. In addition, given that it is attached to the spectacle arm, the data it collects reflect the experience of the eye itself (Wen et al. 2016; Schaeffel 2019; Wen et al. 2019; Wen et al. 2020). Subsequently, a similar device called Vivior has been developed to investigate the visual behaviour patterns of children (Vivior AG, Switzerland) (Mrochen et al. 2020). Additionally, another device called AKESO (Akeso Eyecare Tech Co. Ltd., Beijing, China) that also claims to be able to measure both visual distance and ambient light intensity has been released commercially (www.akeso.cn), but no scientific report has studied it yet. All these portable and wearable devices are valuable tools that researchers can use to investigate the role of these two major features of our current lifestyle that might be associated with myopia.

In this study, we first validated the Clouclip measurements. We then compared the objective results for near-work and outdoor time obtained with the Clouclip to subjective results obtained from a slightly modified version of the questionnaire used in the Guangzhou Outdoor Activity Longitudinal Study (GOALS) (He et al. 2015) to offer a comprehensive report of these two types of essential research tools in the study of myopia.

Methods

The participants in this study were fifth-graders recruited from Lao Liangcang Primary School, which is located in Ningxiang, a county-level city with a population of around 1.4 million near Changsha, Hunan Province, central China. All participants underwent a comprehensive ocular examination, including ocular health and cycloplegic autorefraction. Cycloplegia was induced by instilling three cycles of cyclopentolate 1% (one drop) each 5 min apart. Cycloplegia was then evaluated by testing the light reflex 30 min after the last administration of cyclopentolate. Subjects with incomplete cycloplegia were excluded from the analysis. An autorefractor (model AR-1, Nidek, Aichi, Japan) was used to perform autorefraction. The measurement was repeated three times, and the mean was used for the analysis.

This study complied with the Helsinki Declaration and was approved by the Aier Eye Hospital Group Ethics Committee (No. IRB2016004). Written informed consent was obtained from the students and their parents after an explanation of the study was provided.

Clouclip

The Clouclip device (First generation, Glasson Technology Co. Ltd., Hangzhou, China) has a built-in infra-red distance sensor (measurement range: 15–60 cm) and a light intensity sensor (measurement range: 1–65 336 Lux) for detecting working distance and ambient light intensity in real time (Fig. 1). Clouclip is also equipped with a three-axis accelerometer (X, Y, and Z axes). If the triaxial accelerometer does not detect any change for more than 40 s, the Clouclip enters 'sleep mode' and no data are recorded. Clouclip is otherwise programmed to measure the working distance every 5 s and ambient illuminance every 120 s. It is worn on the right arm of the eyeglass frame to measure the viewing distance and eye-level illuminance approximately along the line of sight. For the participants who did not wear spectacles, frames without lenses were provided so that the Clouclip could be fitted. The participants were asked to wear the Clouclip throughout the day, except while bathing or sleeping, continuously for 1 week (including five weekdays and two weekend days).

Validation study of Clouclip

The distance and light intensity sensors of all the Clouclips used in this study were calibrated by the manufacturer. We also performed a validation test to determine their precision and accuracy. Fifteen Clouclips randomly selected from the 80 used in this study were mounted on a board and were moved through 16 different distances (5–60 cm) along the line of their direction of measurement to evaluate their ability to measure distance. Then, the devices were placed in 28 different light environments (50–38 000 Lux) to compare the readings from Clouclip with those from an illuminometer (model T-10, Konica Minolta, Tokyo, Japan). Five measurements for each Clouclip at the various distances and light intensities were recorded.

Questionnaire

The GOALS questionnaire, with some minor modifications to adapt it to the specific population studied (e.g. adding questions regarding activities that occur during 'morning reading before

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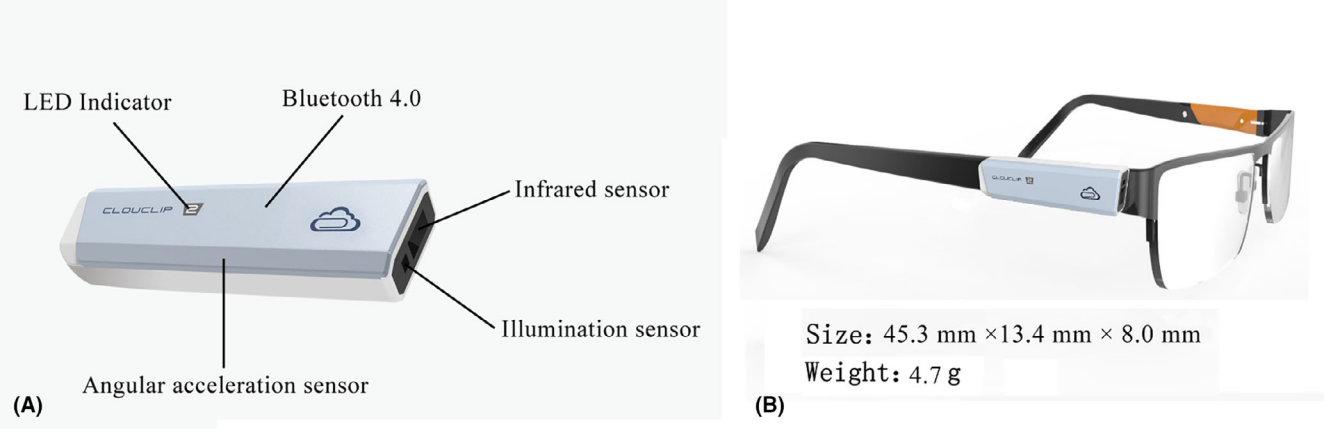


Fig. 1. The Clouclip device. Locations of the sensors on the Clouclip device (A) and Clouclip attached to the right arm of a spectacle frame (B). A downward angle of 10 degrees was set in front of the device.

formal class' and 'lunch break'), was used (Appendix 1). Questions regarding the total amount of time spent on an activity and the type of activities performed were categorized into three distinct periods: school hours, after-school, and the weekend. Activities that were performed outside a building, for example, jogging, playing, or outdoor sports, were classified as outdoor activities; however, outdoor time also included more passive activities such as sitting outside and socializing with friends. Near-work activities included reading, writing, and playing computer or video games.

After 1 week of wearing the Clouclip, all the children were asked to complete the questionnaire to assess the amount of time spent outdoors and in performing near-work activities during that week. Initially, the children completed the questionnaire by themselves (unguided questionnaire). They then filled it in again with guidance from the research team (guided questionnaire).

Data analysis

For the unguided and guided responses to the questionnaire, the average daily amount of time spent outdoors and in performing near-work activities were calculated using the total time recorded from all the questions for each period regarding outdoor time and near-work activities.

Clouclip provides objective measurements of viewing distance and light exposure. For questionnaires, however, only estimates of the time spent on specific classes of activities can be provided, making comparisons between the estimates more difficult. Therefore, we defined cut-off distances that would likely correspond to near-work activities. Most activities classified as near-work are conducted with a viewing distance of <50 cm, but lower cut-offs of <40 cm and <30 cm have also been reported. Based on other studies (Dharani et al. 2012; Alvarez & Wildsoet, 2013), we used a cut-off light intensity of 1000 Lux to distinguish indoor from outdoor environments. Since the Clouclip can only measure a maximum distance of 60 cm, time spent on activities at a working distance that exceeded this limit, such as watching television, was not be measured.

Data from Clouclip were categorized into the school period (7:00 AM to 3:30 PM, 8.5 hr) and the after-school period (3:30–8:00 PM, 4.5 hr) on the weekdays, but were categorized as only one period (7:00 AM to 8:00 PM, 13 hr) on the weekends. Only when data were obtained from ≥80% of the total required wearing time during school hours and ≥60% of the required wearing time during the after-school period and weekend were they considered valid. A valid subject sample was identified as one with a valid period of at least two days during the week and at least one day during the weekend.

Statistical analysis

Data with normal distributions are expressed as the mean ± standard deviation ($M \pm SD$). Data with non-normal distributions are expressed as the median (25th percentile, 75th percentile). Statistical analyses were performed using SPSS 25.0 (SPSS Inc., Chicago, IL, USA). The cut-off for statistical significance was set at $p < 0.05$. For the validation of Clouclip, intra- and inter-class coefficients were used to assess precision. A one sample *t*-test and Bland–Altman analysis were used to evaluate accuracy. The rank sum test and Spearman's correlation analysis were used to compare the answers in the guided and unguided questionnaires regarding the amount of time spent outdoors and in performing near-work activities. The objective data on the time spent exposed to various light levels and working distances obtained from Clouclip were compared to the recorded time spent outdoors or doing near-work from the guided questionnaires using the rank sum test, Bland–Altman analysis, and Spearman's correlation analysis.

Results

General characteristics

A total of 80 students (aged 11.01 ± 0.41 years) were recruited. All the students completed the questionnaires and 74 (92.5%) underwent cycloplegic refraction. The prevalence of myopia, defined as a spherical equivalent of ≤−0.5 D, was 43.8%, with a mean spherical equivalent of −1.04 ± 1.45 D (range, −4.94 to +1.00 D). Clouclip was worn by 64 (80.0%), 48 (60.0%), and 41 (51.3%) out of the 80 students for periods of time that met the data validity standard during the school period, after-school period, and weekend period, respectively (Table 1). No differences

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Table 1. Time the Clouclip was worn during different periods of the study.

	N	Age*	Dioptre*	Percentage f emales	Mean valid days (days)*	Mean daily wearing time (hr)*
School period (7:00 AM to 3:30 PM, 8.5 hr)	64	10.98 ± 0.38	−1.05 ± 1.47	56.3%	4.33 ± 076	8.11 ± 0.21
After-school period (3:30–8:00 PM, 4.5 hr)	48	10.96 ± 0.41	−1.31 ± 1.39	56.3%	3.31 ± 0.78	4.09 ± 0.36
Weekend (7:00 AM to 8:00 PM, 12.5 hr)	41	10.98 ± 0.42	−1.50 ± 1.68	53.7%	1.63 ± 0.49	10.59 ± 1.26

* Data were shown as mean ± standard error.

were found in terms of age, sex ratio, or refractive error between the retained and excluded subjects.

Validation of the Clouclip device

Clouclip provided high-precision measurements of distance and light intensity. The intra-Clouclip correlation coefficients for both parameters for the five repeat measurements were 1.00. The coefficient of variance (CV) for the sixteen measured distances were all <0.91%, except for the CV for 5 cm (2.57%) and 10 cm (1.29%). The CV for the 28 measured light intensities were all <0.15%, except for the CV for 38 000 Lux (0.45%). The inter-Clouclip correlation coefficients for distance and light intensity were 0.99 and 1.00, respectively.

In terms of accuracy, although there were statistically significant differences between the measured values from Clouclip and the true values at different levels of light intensities and distances, the differences were very small. For distance, the maximum error, which occurred at 60 cm, was 1.24 cm with a relative error of 2.07%. For light intensity, the maximum error, which occurred at 38 000 Lux, was 847.04 Lux, with a relative error of 2.23%. The Bland–Altman plots showed a narrow 95% limit of agreement (95% LoA) for both distance and light intensity (Fig. 2). These findings indicate that Clouclip has good accuracy for measuring distance and light intensity. Details of the results for the different measured distances and light intensities are presented in Appendix 2 (Tables A1 and A2).

Comparison of data from guided and unguided questionnaires

In the correlation analysis, the near-work time obtained from the guided and unguided questionnaires was very consistent for the school period, after-school period, and the weekend ($r = 0.65$, $p < 0.01$; $r = 0.85$, $p < 0.01$; $r = 0.80$, $p < 0.01$, respectively). In the rank sum test, there were no differences in near-work time obtained from the guided and unguided questionnaires for the three analysis periods (Table 2). A significant correlation was found between the guided and unguided questionnaires with regard to the time spent outdoors during the school period and the after-school period ($r = 0.71$, $p < 0.01$; $r = 0.79$, $p < 0.01$, respectively). However, for these two different questionnaire approaches, outdoor time on the weekend was less strongly correlated ($r = 0.33$, $p < 0.01$). In the rank sum test, as with near-work time, the outdoor time recorded in the guided questionnaires did not differ from that recorded in the unguided questionnaires (Table 2).

Comparison of data from guided questionnaires and Clouclip exposure time at various working distances

Near-work (reading, writing, etc.) for children of this age typically takes place at distances of 20–40 cm but certainly <50 cm (Wang et al. 2013; Bao et al. 2015). As shown in Table 3, when the questionnaire estimates were

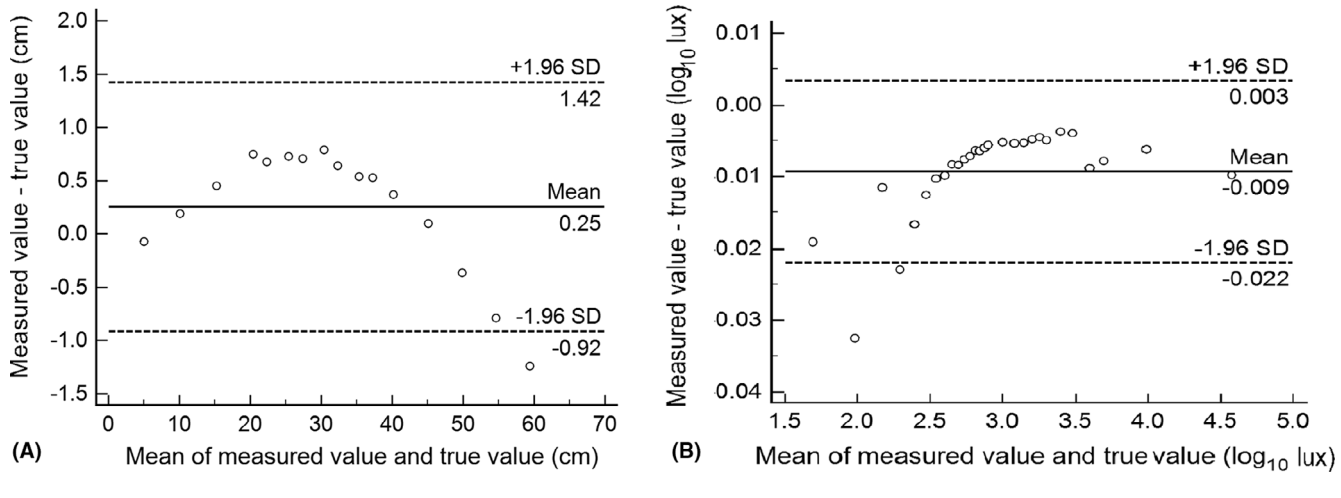


Fig. 2. Agreement between values measured by Clouclip and the true value for distance (A) and light intensity (B) measurements. Solid lines represent the bias between the measured values and true values and dotted lines represent the 95% limit of agreement.

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Table 2. Near-work and outdoor time from the guided questionnaires and unguided questionnaires in different periods.

		Guided questionnaires*	Unguided questionnaires*	p
School period (hr)	Near-work	4.73 (4.73, 5.07)	4.73 (4.46, 4.98)	0.45
	Outdoor exposure	1.60 (1.33, 2.53)	1.60 (1.28, 1.68)	0.12
After-school period (hr)	Near-work	2.00 (1.21, 3.00)	2.00 (1.04, 3.43)	0.94
	Outdoor exposure	0.67 (0.09, 1.14)	0.83 (0.33, 1.33)	0.22
Weekend (hr)	Near-work	2.90 (1.59, 4.78)	2.67 (1.57, 3.98)	0.74
	Outdoor exposure	0.51 (0.02, 1.50)	0.50 (0.00, 1.25)	0.72

* Data were shown as median (25th percentile, 75th percentile).

Table 3. Time spent with various working distances from guided questionnaires and Clouclip in different periods.

	Guided questionnaires*	Clouclip*	p
School period			
Near-work time (hr)	4.73 (4.73, 5.07)	**2.16 (1.74, 2.78) (<30 cm)**	**<0.01**†
		2.88 (2.46, 3.48) (<40 cm)	<0.01†
		3.47 (3.11, 4.10) (<50 cm)	<0.01†
		4.00 (3.57, 4.58) (<60 cm)	<0.01†
After-school period			
Near-work time (hr)	1.46 (0.87, 1.98)	**0.74 (0.60, 1.11) (<30 cm)**	**<0.01**†
		1.04 (0.91, 1.41) (<40 cm)	0.10
		1.35 (1.15,1.78) (<50 cm)	0.53
		1.58 (1.32, 2.11) (<60 cm)	<0.05†
Weekend			
Near-work time (hr)	1.75 (1.00, 2.84)	**1.44 (0.80, 2.95) (<30 cm)**	**0.61**
		2.38 (1.30, 3.70) (<40 cm)	0.14
		2.99 (1.76, 4.55) (<50 cm)	<0.01†
		3.24 (2.00, 4.91) (<60 cm)	<0.01†

* Statistically significant ($p < 0.05$).
† Data were shown as median (25th percentile, 75th percentile).
Bold values represent the time spent in near-work activities with typical distance (<30 cm).

assumed to represent the time spent looking at objects <30 cm away, a typical reading distance, questionnaires had a general tendency to overestimate near-work, regardless of the periods studied. The greatest overestimation occurred during the school period, with an error of 2.57 hr or 2.19 times the measurements from the Clouclip [4.73 hr (4.73, 5.07) versus 2.16 hr (1.74, 2.78); $p < 0.01$]. In addition to the distance of 30 cm, the differences between these two methods were also compared using other distances. During the school period, the questionnaire tended to overrate the duration of near-work, regardless of the cut-offs used.

The results of the comparisons became more complex when children left school. During the after-school period, the exposure time for near-work according to the questionnaires was slightly but statistically lower than that according to Clouclip for the level of distance <60 cm [1.46 hr (0.87, 1.98) versus 1.58 hr (1.32, 2.11); $p < 0.05$], but was similar to the Clouclip estimate when the cut-off was set at <50 cm [1.35 hr (1.15, 1.78)] and at <40 cm [1.04 hr (0.91, 1.41); all $p > 0.05$]. On weekend days, the estimate acquired from the questionnaires [1.75 hr (1.00, 2.84)] was significantly lower than that acquired from the Clouclip for the levels <60 cm [3.24 hr (2.00, 4.91); $p < 0.01$] and <50 cm [2.99 hr (1.76, 4.55); $p < 0.01$]. In contrast, no difference was found between these two approaches [1.75 hr (1.00, 2.84) versus 2.38 hr (1.30, 3.70); $p = 0.14$] for distances <40 cm.

Exposure time at various light levels

The questionnaire could only be used to obtain an estimate of time spent outdoors and could not be used to estimate light intensities. On the assumption that the difference between indoors and outdoors corresponded to a 1000 Lux cut-off, there was an overestimation of the time spent outdoors during the school period in the questionnaire compared to the Clouclip [1.60 hr (1.33, 1.85) versus 1.21 hr (0.96, 1.50); $p < 0.01$]. This overestimation in the questionnaire compared to the Clouclip was also found in the after-school period [0.67 hr (0.33, 1.31) versus 0.34 hr (0.20, 0.58); $p < 0.01$] and on the weekend [0.52 hr (0.15, 1.62) versus 0.17 hr (0.05, 0.39); $p < 0.01$; Table 4].

In addition to the 1000 Lux cut-off, 600, 800, and 1500 Lux were also set as cut-off values for the Clouclip results to differentiate between indoor and outdoor environments (Table 4). During the school period, the estimated time spent outdoors obtained from the questionnaires was slightly but significantly lower than that obtained from the Clouclip [1.60 hr (1.33, 1.85) versus 1.95 hr (1.52, 2.28); $p < 0.01$] when the 600 Lux cut-off was used. However, when the 800 Lux cut-off was used, the outdoor exposure time decreased to 1.44 hr (1.13, 1.83), and no significant difference was found between the approaches ($p = 0.27$). However, when the strictest definition was applied, 1500 Lux, the exposure time was further reduced to 0.89 hr (0.73, 1.07) and the questionnaire results showed a significant overestimation of exposure time ($p < 0.01$).

When the 600 Lux cut-off was used, the estimates from the questionnaires were not significantly different from the time recorded by Clouclip for the after-school period and the weekend ($p = 0.06$ and $p = 0.14$, respectively). However, with a cut-off of 800 or 1500 Lux, an overestimation of exposure time was seen in the questionnaire results compared with the Clouclip (all $p < 0.05$).

Bland–Altman and correlation analysis outcomes between guided questionnaires and Clouclip

We compared the questionnaire estimates with the Clouclip measurements

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Table 4. Time spent with various light levels from guided questionnaires and Clouclip in different periods.

	Guided questionnaires*	Clouclip*	p
School period			
Outdoor time (hr)	1.60 (1.33, 1.85)	1.95 (1.52, 2.28) (>600 lux)	<0.01†
		1.44 (1.13, 1.83) (>800 lux)	0.27
		1.21 (0.96, 1.50) (>1000 lux)	**<0.01†**
		0.89 (0.73, 1.07) (>1500 lux)	<0.01†
After-school period			
Outdoor time (hr)	0.67(0.33, 1.31)	0.45 (0.25, 0.76) (>600 lux)	0.06
		0.39 (0.22, 0.68) (>800 lux)	0.02†
		0.34 (0.20, 0.58) (>1000 lux)	**<0.01†**
		0.27 (0.17, 0.41) (>1500 lux)	<0.01†
Weekend			
Outdoor time (hr)	0.52(0.15, 1.62)	0.30 (0.16, 0.80) (>600 lux)	0.14
		0.23 (0.09, 0.53) (>800 lux)	0.02†
		0.17 (0.05, 0.39) (>1000 lux)	**<0.01†**
		0.07 (0.02, 0.21) (>1500 lux)	<0.01†

* Data were shown as median (25th percentile, 75th percentile).
† Statistically significant ($p < 0.05$).
Bold values represent the time spent in outdoor environments with typical cut-off light intensity (>1000 lux).

using the reasonable assumption that subjective estimates of near-work are likely to correspond to objective measurements of activities performed at viewing distances of <30 cm (Wang et al. 2013; Bao et al. 2015), while subjective estimates of time outdoors are likely to correspond to objectively measured illuminances of >1000 Lux (Fig. 3). It was observed that the width of the 95% LoA for both near-work time and outdoor time were relatively wide, especially during the weekend period. This suggests that the agreement between these two approaches was relatively poor and that the agreement also varied between the periods observed.

Figure 4 shows correlation plots between the results obtained from the questionnaires and those obtained from Clouclip. For near-work time, a weak correlation was found between the questionnaire estimates and Clouclip estimates during the school period and on the weekend ($r = -0.26$, $p = 0.04$; $r = 0.36$, $p = 0.02$, respectively), but no significant correlation was found during the after-school period ($r = 0.28$, $p = 0.06$). With regard to outdoor time, there was a weak correlation between the questionnaire estimates and the Clouclip estimates during the school period ($r = 0.28$, $p = 0.03$); however, the correlation was not significant during the after-school period and on the weekend ($r = 0.26$, $p = 0.08$; $r = 0.16$, $p = 0.32$, respectively).

Discussion

In order to precisely define modifiable risk factors for the development of myopia, these factors must be measured accurately (French et al. 2013a). In this study, we demonstrated that Clouclip, a wearable device that can measure both viewing distance and light exposure, has excellent precision and accuracy. We further compared the results obtained from a traditional subjective questionnaire to those obtained objectively using the Clouclip device. We found that the accuracy of the questionnaires varied depending on the observation period (e.g. weekdays versus weekends) and the type of activity (e.g. near-work within 30 or 40 cm), but under typical assumptions (e.g. near-work <30 cm and light intensity >1000 Lux), questionnaires provided a general tendency towards overestimation of the time spend on near-work and time spent outdoors. Nevertheless, the objective data on time spent outdoors in this study confirms the low amounts of time outdoors reported in a number of studies on children in East Asia (Rose et al. 2008b; Guo et al. 2013). On school days, using a cut-off of >1000 Lux, the total time outdoors was estimated to be 1.55 hr, only 0.34 hr of which occurred after school. This fits with the very low levels reported in questionnaires, which often ask for estimates of time outdoors outside of school hours. It is striking that very low measurements were also obtained for weekend days, suggesting that indoor-oriented lifestyles persist for Chinese children even on the weekends. Higher questionnaire estimates have been obtained for children in Western countries (French et al. 2013b), and some objective data have also confirmed this (Read et al. 2014; Ostrin 2015).

Questionnaires, the classical method to quantify modifiable risk factors of myopia, have been utilized for several decades (Saw et al. 1999) despite their numerous drawbacks, including the variety of questionnaire formats, imprecise qualitative descriptions, and recall bias (Jones-Jordan et al. 2012). Surprisingly, the present study showed good agreement between the guided and unguided questionnaires. One possible reason could be that in the current study, subjects were required to complete the questionnaire immediately after what happened in the last week. Therefore, recall bias may have been minimized. More importantly, the questionnaire used has been revised several times and was further modified to fit the scenario of the current study. Another possible reason is that for students, activities that occur during the school period (e.g. having classes for all subjects) are more uniform than those that occur after school and, especially, during weekends. Therefore, the exposure time of near-work and outdoor activities from the guided and unguided questionnaires tended to be

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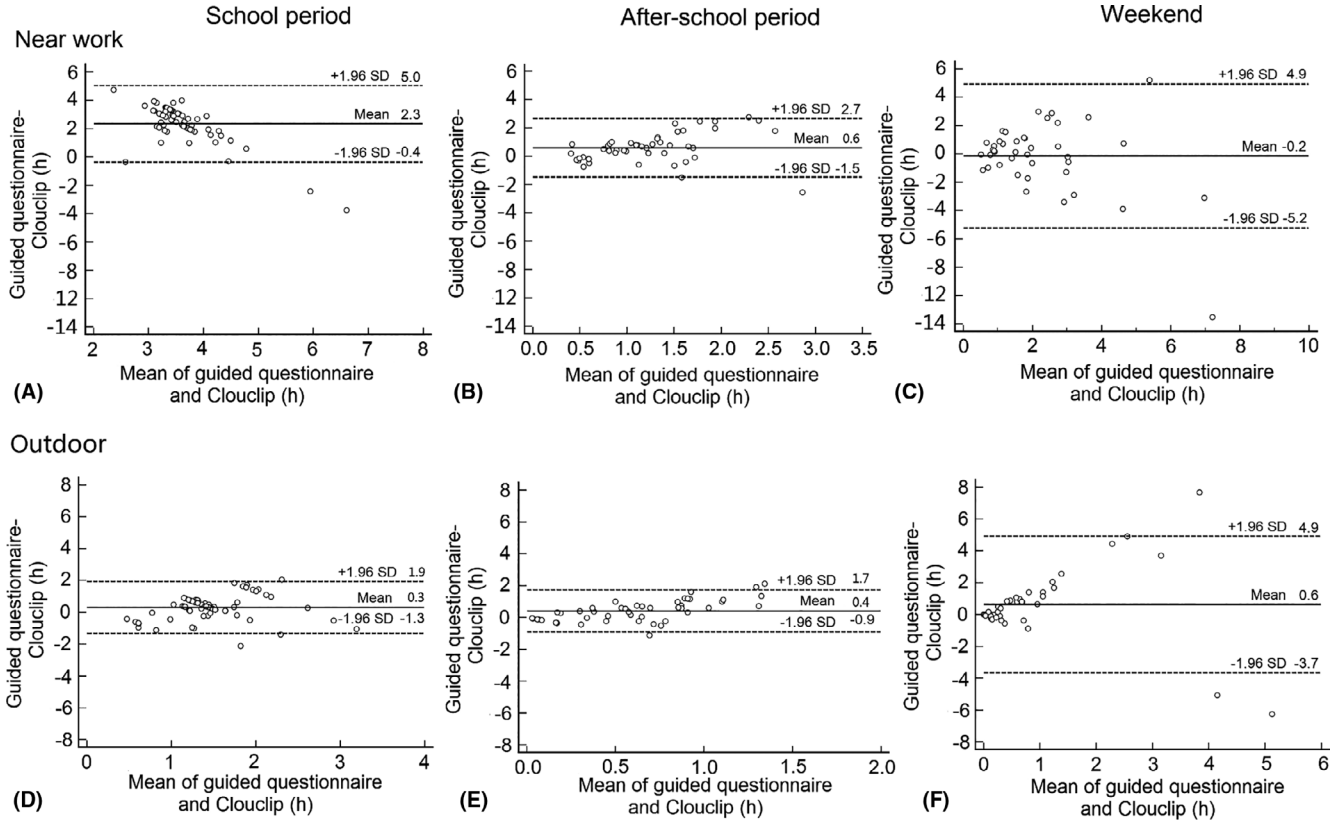


Fig. 3. Agreement of near-work time (A–C) and outdoor time (D–F) between Clouclip and questionnaire results for the different periods. The cut-off value of near-work and outdoor environment for Clouclip was set <30 cm and >1000 Lux, respectively. The solid line indicates the mean difference (bias). The upper and lower lines represent the 95% limit of agreement (LoA).

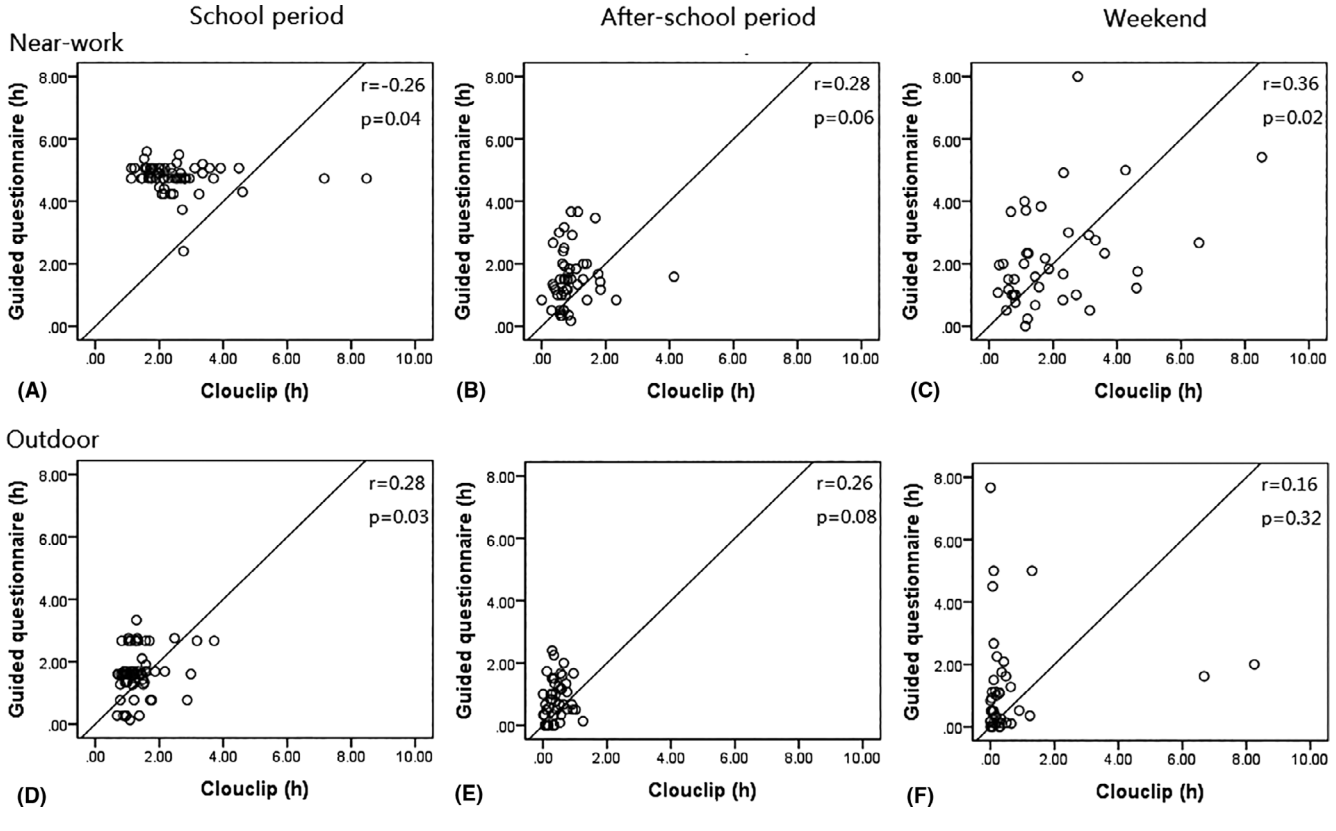


Fig. 4. Correlation plots of near-work time (A–C) and outdoor time (D–F) between guided questionnaire and Clouclip results for the different periods. The cut-off value of near-work and outdoor environment for Clouclip was set <30 cm and >1000 Lux, respectively.

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identical during the school period but tended to differ when the participants were outside of the school environment. This suggests that the questionnaire can provide more reliable estimates when activities are standardized. At the same time, it is important to mention that the time spent outdoors reflected in the questionnaires was very short after school and even on the weekend. A similar amount of time outdoors was also observed in Singapore children (Read et al. 2018). In contrast, dramatically greater exposure time outdoors has been reported in Australian students (French et al. 2013b) and American students (Jones et al. 2007). These different lifestyles might be associated with the differences in myopia prevalence reported in these locations and groups (Rose et al. 2008b).

Owing to recent advances in technology, several wearable devices have been developed to measure the intensity and duration of light exposure, such as HOBO, Acitwatch, and Fitsight. In a study by Schmid et al. (2013), HOBO, a light metre that hangs in front of the chest, was used to quantify daily illuminance to compare indoor illumination and daily sunlight in emmetropic and myopic university students and no significant between-group differences were found. Read et al. (2015) utilized Actiwatch, a light metre in the form of a wristwatch, to examine the relationship between ambient light exposure and axial changes and observed that greater daily light exposure was associated with less axial eye growth. Fitsight, another wristwatch-like light metre that was recently developed, aims to not only record light intensity and/or physical activity, but also to encourage children to increase time spent outdoors (Verkicharla et al. 2017). However, none of these devices have measured near-viewing distance. Additionally, even for light intensity, the values found in these studies were different from those perceived by the eye because the positions of the devices were not in line with the subject's eyesight. In contrast, Clouclip provides a more comprehensive and direct quantification of these risk factors.

Questionnaires attempt to estimate the total duration of time spent outdoors and in doing near-work, while Clouclip measures the time exposed to different light intensities and working distances. To facilitate the comparison between these two methods, we used assumed levels as the thresholds to represent typical scenarios for near-work and outdoor environments. Assuming that 20–30 cm is a typical range of distance for near-work (Wang et al. 2013; Bao et al. 2015), the time reported by children in a questionnaire should have corresponded to Clouclip readings at those distances. Nevertheless, we found that there was a significant difference between the results from these two methods and that the questionnaires consistently overestimated the duration compared with the Clouclip. This suggests that during near-work activity, a significant amount of time may be spent looking up or around at greater viewing distances rather than concentrating only on the near task. However, it seems likely that some students concentrate on the task, maintaining a relatively constant viewing distance, while others may gaze away more often, minimizing the actual 'near-work' exposure. A video of two students with different reading styles is provided in Video S1. However, all students may estimate a similar time spent on a near-work task, despite the differences in viewing patterns. Different viewing patterns have actually, at least in animals, been shown to have a high impact on the development of myopia (Winawer & Wallman, 2002; Kee et al. 2007; Zhu & Wallman, 2009). Therefore, Clouclip, given its capability to provide real-time measurements of distance and detailed temporal patterns of near-work activity, has the potential to provide further information on this issue in humans.

It is interesting to observe that, regardless of the cut-off distances for near-work, during weekdays (including the school period and after school), questionnaire tended to overestimate the duration (with the exception of the after-school period at <60 cm), while for weekend days, questionnaire tended to underestimate the duration. This may be because students perceived 'having class (during school period)' and 'doing homework' (after-school period) as near-work and responded to the questionnaire with the duration for these kinds of near-work activities, whereas Clouclip measured the distance for near-work activities that occurred only within its measurement limit. In contrast, students on weekend days performed many diverse types of near-work activities (e.g. painting, playing the piano, playing hand-held video games, etc.). The complexity of the types of activities and the more fragmented time spent on them in a day compared to the regular, structured classes during the school period may have led subjects to underestimate exposure time for near-work activities.

When 1000 Lux was set as the cut-off value for outdoor exposure, questionnaire overestimated the duration compared with Clouclip, regardless of the periods observed in the study. This was also the case for other cut-off values and for all three periods, except for the 600 Lux cut-off during the school period. This is because the questionnaire asked about the duration of time outside of the building, while Clouclip measured the ambient illuminance. This is especially true for a device such as Clouclip, which aligns with the line of sight rather than facing the sky. The position of the device, or more precisely, the direction of the embedded light sensor, has a significant impact on the results. Although a previous study (Okudaira et al. 1983) reported a correlation coefficient of 0.76 between the values detected at the forehead and wrist, the range of the ambient illuminance measured was <2500 Lux. In fact, we have found that (unpublished data) the variance of the results given by the light sensors placed in different positions increased with ambient illuminance. In an environment with an illuminance of approximately 23 000 Lux (face to the sky), the difference between positions was as high as 13 000 Lux [i.e. line of sight: 15 000 Lux (face slightly upward), chest: 4600 Lux, wrist: 2300 Lux]. Similar to viewing distance, the light intensity perceived by the eye is heavily influenced by the line of sight (Fig. 5). In addition, the temporal pattern of the variation in illuminance was also found to affect myopia development experimentally (Lan et al. 2014). Since illuminance is accepted as a factor that directly impacts the development of myopia, obtaining both the level and the temporal pattern of light intensity perceived by the eye should be a key concern for future studies.

Several papers have compared subjective methods with objective methods to quantify myopia-related behaviours.

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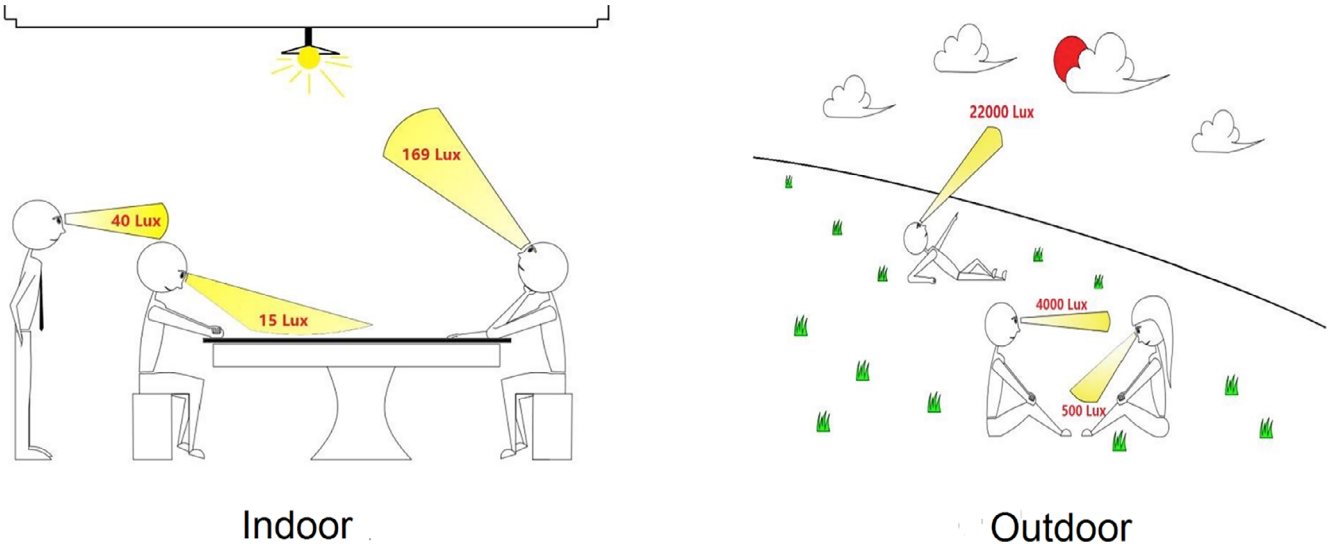


Fig. 5. Illustration of different levels of light intensities perceived by the eye along different lines of sight in the same indoor or outdoor environment.

Dharani et al. (2012) compared time outdoors obtained from a visual activity diary with that obtained from HOBO, using a light level threshold of 1000 Lux for outdoor activity. They also reported poor agreement between subjective and objective methods. In addition, Alvarez & Wildsoet (2013) also compared the amount of time spent outdoors obtained from questionnaires with that obtained from HOBO, and found that the questionnaire responses provided a general overestimation across seasons, which was similar to our results. Further, Ostrin (2017) compared the time spent outdoors obtained from a questionnaire with that obtained from another objective device, Actiwatch, and also found that in the questionnaire, respondents overestimated the time spent outdoors compared to the Actiwatch data. Although all these studies have found a significant difference between questionnaire estimates and the data obtained from objective devices, this does not mean that questionnaires should be abandoned. One advantage of questionnaires is their low cost, which makes them particularly suitable for large-scale (Lee et al. 2013) and long-term (Deng et al. 2010) studies. Many questionnaires ask for estimates of typical patterns of activity, and better agreement between subjective and objective measures may be obtained when questionnaires ask about activity patterns for the same period over which objective measures are obtained. If this is the case, a better approach to reliable questionnaire estimates may be obtained by experience sampling-based approaches, such as daily diaries. In addition, the reliability of the responses depends heavily on the design of the questionnaire itself and the gap between when the activities occur and the time the questionnaire is completed. It is especially suitable to estimate the duration of modularized activities. On the other hand, although objective measurements may provide accurate and direct measurements of the metrics of interest, they are only suitable for studies with relatively small sample sizes.

In summary, Clouclip is a novel and objective device that can be used to document both light intensity and working distance. The agreement between the questionnaires and Clouclip measurements of these two risk factors in this study varied depending on the period and the type of activity being queried. However, given their strengths, the Clouclip and questionnaires can be used to complement each other to provide a more logical and feasible assessment of modifiable risk factors for myopia. However, the objective measures obtained with the Clouclip device in this study confirm the low subjective estimates of time outdoors found in several studies on East Asian children.

Data Availability Statement

The data used to support the findings of this study are available from the corresponding author (Weizhong Lan, Zhikuan Yang) upon request.

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Appendix 1

Children's daily activity questionnaire (completed by children)

Name of School: **Grade:** **Class:** **Name:**
Date of birth: **(MM/DD/YYYY)** **Sex:**

Please refer to the instruction and sample and fill out the questionnaire as best as you can

Part 1: On a typical school weekday **Please fill out the time you spend doing the following activities**			
(1) Before school			
Activity	**Undertake or not**	**If yes**	
		Time/Duration	**When does it start and end?**
1. When do you get up in the morning?	Yes	At __:__	Not applicable
2. Any outdoor activities (e.g., jogging or playing) before going to school?	□Yes □No	_ Minutes in toal	From__: __ to __:__
3. How do you usually travel to school?	□By school bus or regular bus □By sedan □By bike, motor bicycle, or on foot		
4. How long does it take to travel to school?	Yes	_ minutes in total	From__: __ to __:__
(2) At School			
5. Do you do morning reading?	□Yes □No	_ minutes in total	From___: __ to __:__
6. How many classes do you have in a **normal weekday**?	Yes	_ classes in total	Not applicable
7. How long is the break time between classes **in a normal weekday**?	Yes	_ mins in total	Not applicable
8. Do you prefer to **stay in the classroom** or **play outside the classroom** during the breaks?	□ Stay in the classroom □ Play outside the classroom (like playground)		
9. Do you do setting-up exercises during the break regularly?	□Yes □No	_ mins in total	From___: __ to __:__
(3) Lunch break			
10. How long is the break time between lunch and self-study at noon **in a normal weekday**?			
11. Do you prefer to stay indoors or outdoors during the break?	□ Indoors □ Outdoors (e.g., playground, street)		
12. How long is the self-study at noon?	Yes	_ minutes in total	From__: __ to __:__
(4) After school			
13. How do you usually travel back home?	□By school bus or regular bus □By sedan □By bike, motor bicycle, or on foot		
14. Do you prefer to have outdoor activities (e.g., running, playing, shopping or wandering) after school before night fall?	□Yes □No	_ minutes in total	Part 1: From__: __ to __: Part 2: From__: __ to __:
15. Do you do homework or read books for pleasure?	□Yes □No	_ minutes in total	Part 1: From__: __ to __: Part 2: From__: __ to __:
16. Do you watch TV?	□Yes □No	_ minutes in total	Part 1: From__: __ to __: Part 2: From__: __ to __:
17. Do you play video/computer games?	□Yes □No	_ minutes in total	Part 1: From__: __ to __: Part 2: From__: __ to __:
18. When do you go to bed at night?	Yes	At __:__	Not applicable

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Part 2： Overview of typical school weekdays			
1.How many physical education classes do you have during the five school weekdays?	□Yes □No	___ times per week	Not applicable
2. Does your child take part in any **indoor extracurricular** training classes (e.g., English class, piano class) during the five school weekdays?	□Yes □No	__ minutes in total	Not applicable
3. Does your child take part in any **outdoor extracurricular** training class (e.g., soccer class, basketball class) during the five school weekdays?	□Yes □No	_ minutes in total	Not applicable

Part 3：On a typical day of school weekend (Saturday or Sunday)			
Activity	**Undertake or not**	**If yes**	
		Time/Duration	**When does it start and end?**
1. When do you get up in the morning?	Yes	At __:__	Not applicable
2. Do you take part in any **indoor extracurricular** training classes (e.g., English class, piano class)?	□Yes □No	__ minutes in total	Part 1: From__: __ to __: Part 2: From__: __ to __:
3. Do you take part in any **outdoor extracurricular** training classes (e.g., soccer class, basketball class)?	□Yes □No	__ minutes in total	Part 1: From__: __ to __: Part 2: From__: __ to __:
4. Do you study or read in addition to **the extracurricular training classes and homework**?	□Yes □No	__ minutes in total	Part 1: From__: __ to __: Part 2: From__: __ to __:
5. Do you watch TV?	□Yes □No	__ minutes in total	Part 1: From__: __ to __: Part 2: From__: __ to __:
6. Do you child play video/computer games?	□Yes □No	__ minutes in total	Part 1: From__: __ to __: Part 2: From__: __ to __:
7. In addition to **outdoor extracurricular classes**, are there any other outdoor activities (e.g., exercise, playing, shopping or wandering)?	□Yes □No	__ minutes in total	Part 1: From__: __ to __: Part 2: From__: __ to __:
8. When do you go to bed?	Yes	At __:__	Not applicable

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Appendix 2

Table A1. One-sample *t* test outcomes and CVs for distance measurements obtained using Clouclip.

True value (cm)	Measured value (cm)	Absolute error (cm)	Relative error (%)	CV (%)	*t*	p
5	4.93 ± 0.51	−0.07	−1.40	2.57	1.21	0.23
10	10.19 ± 0.53	0.19	1.90	1.29	−3.01	<0.01*
15	15.45 ± 0.60	0.45	3.00	0.91	−6.46	<0.01*
20	20.75 ± 0.61	0.75	3.75	0.70	−10.60	<0.01*
22	22.68 ± 0.53	0.68	3.09	0.72	−11.16	<0.01*
25	25.73 ± 0.54	0.73	2.92	0.64	−11.68	<0.01*
27	27.71 ± 0.58	0.71	2.63	0.61	−10.47	<0.01*
30	30.79 ± 0.60	0.79	2.63	0.56	−11.40	<0.01*
32	32.64 ± 0.58	0.64	2.00	0.52	−9.54	<0.01*
35	35.54 ± 0.63	0.54	1.54	0.51	−7.43	<0.01*
37	37.53 ± 0.63	0.53	1.43	0.53	−7.31	<0.01*
40	40.37 ± 0.77	0.37	0.92	0.54	−4.16	<0.01*
45	45.10 ± 0.98	0.10	0.22	0.51	−0.87	0.39
50	49.64 ± 1.17	−0.36	−0.72	0.41	2.66	<0.01*
55	54.21 ± 1.44	−0.79	−1.44	0.44	4.76	<0.01*
60	58.76 ± 1.77	−1.24	−2.07	0.45	6.04	<0.01*

CV = coefficient of variation.
* Statistically significant ($p < 0.05$).

Table A2. One-sample *t* test outcomes and CVs for light intensity measurements obtained using Clouclip

True value (lux)	Measured value (lux)	Absolute error (lux)	Relative error (%)	CV (%)	*t*	p
50	47.85 ± 1.79	−2.15	−4.30	0.12	10.38	<0.01
100	92.80 ± 16.93	−7.20	−7.20	0.00	3.68	<0.01*
150	146.07 ± 12.46	−3.93	−2.62	0.00	2.73	<0.01*
200	189.73 ± 7.33	−10.27	−5.14	0.00	12.13	<0.01*
250	240.60 ± 6.58	−9.40	−3.76	0.00	12.36	<0.01*
300	291.47 ± 6.18	−8.53	−2.84	0.00	11.95	<0.01*
350	341.80 ± 6.33	−8.20	−2.34	0.00	11.23	<0.01*
400	391.07 ± 7.48	−8.93	−2.23	0.00	10.34	<0.01*
450	441.47 ± 8.90	−8.53	−1.90	0.00	8.31	<0.01*
500	490.47 ± 8.65	−9.53	−1.91	0.00	9.54	<0.01*
550	540.47 ± 9.73	−9.53	−1.73	0.00	8.49	<0.01*
600	590.20 ± 9.74	−9.80	−1.63	0.00	8.71	<0.01*
650	640.60 ± 11.12	−9.40	−1.45	0.00	11.96	<0.01*
700	689.73 ± 11.09	−10.27	−1.47	0.00	8.02	<0.01*
750	739.80 ± 10.86	−10.20	−1.36	0.00	8.13	<0.01*
800	789.73 ± 10.32	−10.27	−1.28	0.00	8.61	<0.01*
1000	988.07 ± 13.21	−11.93	−1.19	0.00	7.82	<0.01*
1200	1185.20 ± 15.39	−14.80	−1.23	0.00	8.33	<0.01*
1400	1383.13 ± 20.11	−16.87	−1.20	0.00	7.26	<0.01*
1600	1582.47 ± 22.61	−17.53	−1.10	0.00	6.72	<0.01*
1800	1781.53 ± 24.71	−18.47	−1.03	0.00	6.47	<0.01*
2000	1977.33 ± 28.85	−22.67	−1.13	0.00	6.81	<0.01*
2500	2478.67 ± 28.94	−21.33	−0.85	0.00	6.38	<0.01*
3000	2973.00 ± 30.94	−27.00	−0.90	0.00	7.56	<0.01*
4000	3919.60 ± 122.08	−80.40	−2.01	0.00	5.70	<0.01*
5000	4911.33 ± 135.51	−88.67	−1.77	0.00	5.67	<0.01*
9810	9671.72 ± 116.50	−138.28	−1.41	0.15	10.28	<0.01*
38 000	37152.96 ± 680.46	−847.04	−2.23	0.45	10.78	<0.01*

CV = coefficient of variation.
* Statistically significant ($p < 0.05$).

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Supporting Information

Additional Supporting Information may be found in the online version of this article:

Video S1. A video of two students with different reading styles.

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